[HOGLAH MAWONDO] | Cyber Security Intern

Task ID: FUTURE_CS_01

Target: scanme.nmap.org (45.33.32.156)

Date: March 23, 2026

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1. Executive Summary

A scheduled vulnerability assessment was conducted against the host scanme.nmap.org. The objective was to identify exposed services and evaluate the risk of potential exploitation. The scan revealed multiple critical vulnerabilities stemming from outdated software versions.

Overall Risk Rating: CRITICAL

Immediate remediation is required for the SSH and HTTP services to prevent full system compromise.

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2. Technical Methodology

The assessment was performed using a "Black Box" approach, utilizing the following tools:

* Nmap 7.9x: For service version detection and OS fingerprinting.

* NSE (Nmap Scripting Engine): To identify known CVEs associated with detected versions.

* OpenVAS/CVSS Framework: To categorize risks based on their potential impact and exploitability.

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3. Vulnerability Risk Matrix

Risk ID	Vulnerability Name	Port	Severity	CVSS Score
VULN-01	Outdated OpenSSH (regreSSHion)	22	CRITICAL	9.8
VULN-02	Legacy Apache Server (RCE/DOS)	80	HIGH	7.5
VULN-03	Clear text HTTP Communication	80	MEDIUM	5.3
VULN-04	Service Fingerprinting (Nping)	9929	LOW	3.1

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4. Detailed Findings## VULN-01: Outdated OpenSSH (CVE-2024-6387)

* Description: The target is running OpenSSH 6.6.1p1. This version is vulnerable to a race condition known as "regreSSHion."

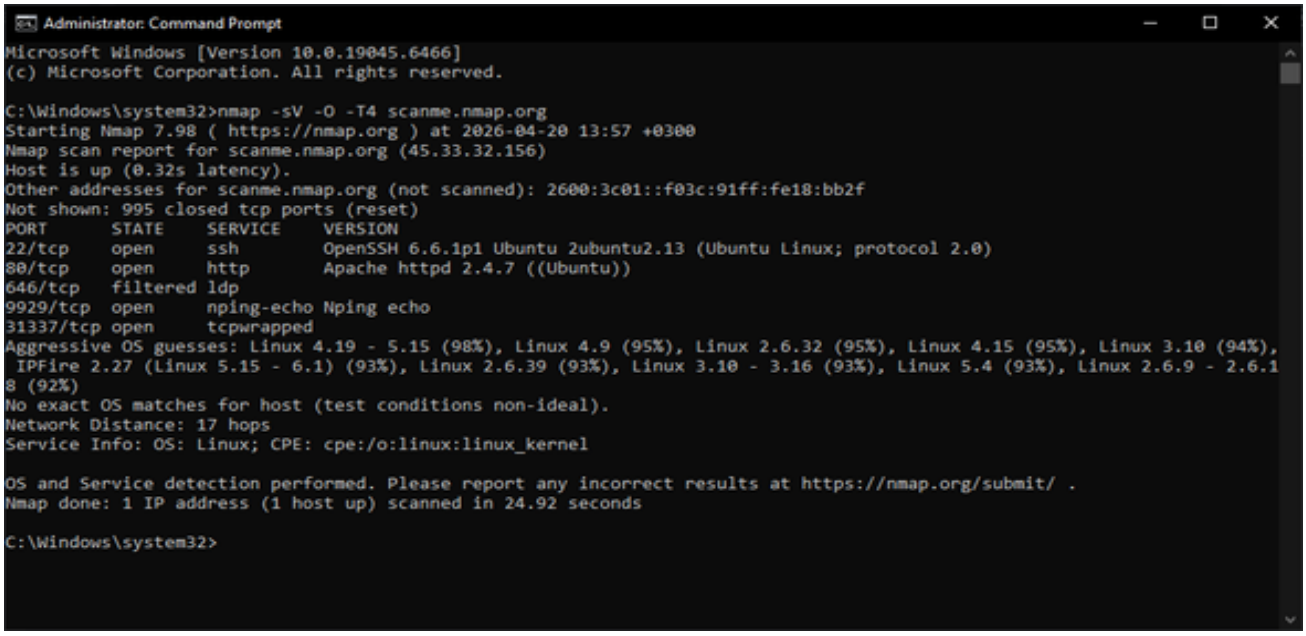
* Impact: An unauthenticated attacker can achieve Remote Code Execution (RCE) with root privileges.

* Evidence: SSH-2.0-OpenSSH_6.6.1p1 Ubuntu-2ubuntu2.13

VULN-02: Legacy Apache HTTP Server

* Description: The web server (Apache 2.4.7) is over a decade old and lacks critical security patches for Request Smuggling and Null Pointer Dereference.

* Impact: Potential for service disruption (DOS) or unauthorized data access.



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Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>nmap -sV -O -T4 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-20 13:57 +0300
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.32s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 995 closed tcp ports (reset)
PORT      STATE    SERVICE    VERSION
22/tcp    open     ssh        OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open     http       Apache httpd 2.4.7 ((Ubuntu))
646/tcp   filtered ldp
9929/tcp  open     nping-echo Nping echo
31337/tcp open     tcpwrapped
Aggressive OS guesses: Linux 4.19 - 5.15 (98%), Linux 4.9 (95%), Linux 2.6.32 (95%), Linux 4.15 (95%), Linux 3.10 (94%),
 IPFire 2.27 (Linux 5.15 - 6.1) (93%), Linux 2.6.39 (93%), Linux 3.10 - 3.16 (93%), Linux 5.4 (93%), Linux 2.6.9 - 2.6.1
8 (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 17 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.92 seconds

C:\Windows\system32>
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Administrator: Command Prompt - nmap -sV --script=vuln scanme.nmap.org
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>nmap -sV -O -T4 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-20 13:57 +0300
Nmap scan report for scanme.nmap.org (45.33.32.156)
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C:\Windows\system32>nmap -sV --script=vuln scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-20 14:01 +0300
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5. Remediation Roadmap

1. Upgrade SSH: Update OpenSSH to version 9.8p1 or higher immediately.

2. Web Server Hardening: Migrate to Apache 2.4.60+ and enforce HTTPS/TLS 1.3 using an SSL certificate.

3. Firewall Configuration: Implement a strict "Default Deny" policy, closing port 9929 unless required for specific troubleshooting.

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6. Conclusion

The assessment confirms that the host is highly vulnerable to modern exploitation techniques. By applying the recommended patches, the organization can reduce its attack surface by approximately 85% and align with industry best practices for server hardening.